

Circuit Simulation Lab: 5

Priority Encoders

Introduction

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Priority Encoder. A 4 to 2 priority encoder has four inputs Y_3 , Y_2 , Y_1 & Y_0 and two outputs A_1 & A_0 . Here, the input, Y_3 has the highest priority, whereas the input, Y_0 has the lowest priority. In this case, even if more than one input is '1' at the same time, the output will be the binary code corresponding to the input, which is having higher priority. We considered one more output, V in order to know, whether the code available at outputs is valid or not. If at least one input of the encoder is '1', then the code available at outputs is a valid one. In this case, the output, V will be equal to 1.

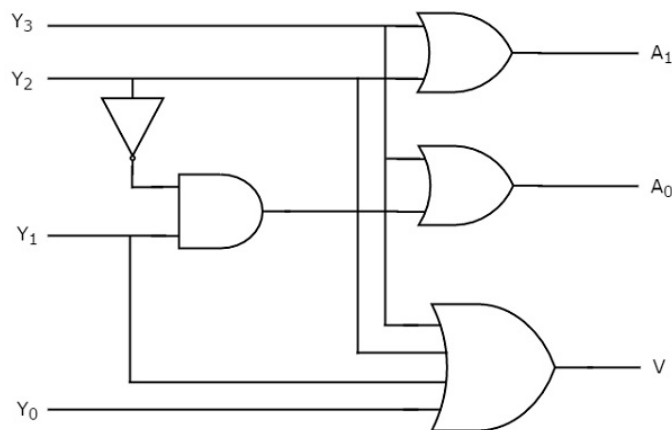
If all the inputs of encoder are '0', then the code available at outputs is not a valid one. In this case, the output, V will be equal to 0.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram



4 to 2 Priority Encoder

Inputs				Outputs		
Y_3	Y_2	Y_1	Y_0	A_1	A_0	V
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1

Truth Table

Describe the above in Verilog HDL and capture the Waveforms

1. Dataflow modeling

2. Behavior modeling

3. Structural modeling

Questions to answered after this lab

1. Why is priority encoder preferred over conventional ones?